

Capstone Detailed Outline: Dynamic Benchmarking of LLMs: Moving Beyond Static Evaluation

I. Introduction and Background

A. The Rapid Advancement and Deployment of Large Language Models (LLMs)

- **Context:** Large Language Models (LLMs) have become foundational to language technologies, but standardizing their evaluation remains a critical challenge.
- **The Evaluation Challenge:** As models scale, benchmarking efforts must continuously adapt; holistic evaluations are necessary to assess models across a broad range of scenarios, metrics, and real-world applications (Liang et al., 2022).

B. The Vulnerability of Static Evaluation

- **Defining the Baseline:** Standardized tests were originally designed to measure broad knowledge acquired during pretraining, testing subjects ranging from the humanities to STEM (Hendrycks et al., 2021).
- **The Data Contamination Problem:** As models ingest massive portions of the internet, static benchmarks face a high risk of test set leakage. Researchers note that "test set contamination, wherein test data from a benchmark ends up in a newer model's training set, is a well-documented obstacle for fair LLM evaluation and can quickly render benchmarks obsolete" (White et al., 2024, p. 1). High scores may therefore reflect statistical familiarity with the training data rather than generalized reasoning.

C. Thesis and Research Question

- **Research Question:** Compared with static evaluation benchmarks like MMLU, how do dynamic evaluations (LiveBench) and formal metalinguistic evaluations (the Beguš et al. linguistic dataset) alter the performance rankings and assessed reasoning capabilities of contemporary LLMs?
- **Hypothesis:** While frontier models will continue to outperform baseline models across all formats, the performance gap will shrink under dynamic and metalinguistic conditions where models cannot rely on data contamination.

II. Literature Review: Assessing the Evaluation Frameworks

A. Traditional Static Benchmarking (MMLU & GLUE)

- **GLUE:** Early natural language understanding evaluations like GLUE were designed with the goal "that good performance should require models to share substantial knowledge (e.g., trained parameters) across tasks" (Wang et al., 2018, pp. 353-354).
- **MMLU:** The Measuring Massive Multitask Language Understanding (MMLU) benchmark evaluates multi-task accuracy across 57 subjects to analyze aggregate properties and identify blind spots (Hendrycks et al., 2021). However, studies show that "state-of-the-art models still struggle at learning and applying knowledge from pretraining" and exhibit "lopsided performance and frequently do not know when they are wrong" (Hendrycks et al., 2021, pp. 1-2).

B. The Shift Toward Dynamic Benchmarks

- **Dynamic Solutions (LiveBench):** To mitigate test set contamination, dynamic benchmarks like LiveBench have been developed because they contain "frequently-updated questions from recent information sources" and score answers "automatically according to objective ground-truth values" (White et al., 2024, p. 1).
- **Preventing Memorization:** LiveBench relies on a diverse set of categories, including recent math competitions, newly published arXiv papers, and current news articles, to ensure that the questions are novel and untainted by previous training datasets (White et al., 2024).

C. Evaluating Metalinguistic Abilities in LLMs

- **The Linguistics Gap:** Most benchmarks focus on general knowledge, but computational linguistics requires models to reason about language itself. Beguš et al. (2025) define this as "metalinguistic abilities, i.e., with abilities not only to use language, but also to explicitly reason about it" (p. 3454).
- **Current Metalinguistic Performance:** Recent studies testing models on syntax and phonology indicate that OpenAI o1 "vastly outperforms other models on a variety of linguistic analysis tasks, being able to generate correct syntactic and phonological analyses" (Beguš et al., 2025, p. 3454). This challenges earlier claims that language models "lack the skill of iterative reasoning upon knowledge" (Şahin et al., 2020, as cited in Beguš et al., 2025, p. 3464).

Here are the fully updated Methodology and Bibliography sections for your outline. I have seamlessly integrated the official, primary-source APA citations for GPT-5.4, Claude Sonnet 4.6, and MiniMax M2.7 directly into the text.

III. Methodology

(Note: Designed to be executable within a highly constrained timeframe, utilizing one static, one dynamic, and one existing linguistic dataset).

A. Selected Models and Setup

- This study will evaluate **three top-of-the-line frontier models, all released in early 2026**, capturing the most current state of AI capabilities while optimizing for budget and inference speed: OpenAI's **GPT-5.4** (released March 5, 2026), Anthropic's **Claude Sonnet 4.6**, and MiniMax's **M2.7** (Anthropic, 2026; MiniMax, 2026; OpenAI, 2026).
- These specific models were selected to provide a balanced analysis across proprietary architectures and highly efficient models. GPT-5.4 serves as the flagship, general-purpose baseline, bringing together advances in reasoning, coding, and agentic computer-use capabilities into a single model (OpenAI, 2026). Claude Sonnet 4.6 is included as a highly capable model that substantially improves upon its predecessor to approach the capability levels of the frontier Opus 4.6 model, while remaining cost-effective (Anthropic, 2026). MiniMax M2.7 is included as an extremely efficient frontier alternative that acts as the first model deeply participating in its own self-evolution (MiniMax, 2026).
- All models will be tested under uniform conditions. As noted in recent linguistic evaluation studies, "within the range of 0.0–1.0, changes in temperature do not affect the LLMs' problem-solving ability" (Beguš et al., 2025, p. 3456).

B. Benchmark 1: Static Evaluation (MMLU)

- Evaluate models on a subset of the MMLU benchmark, which includes "57 tasks including elementary mathematics, US history, computer science, law, and more" (Hendrycks et al., 2021, p. 1).
- This establishes the traditional, potentially contaminated baseline ranking.

C. Benchmark 2: Dynamic Evaluation (LiveBench)

- Execute models on the reasoning and language comprehension tasks from LiveBench, which "contains a wide variety of challenging tasks, spanning math, coding, reasoning, language, instruction following, and data analysis" (White et al., 2024, p. 1).
- This tests the models' ability to handle novel problems that have not been memorized during pretraining by relying on frequently-updated questions from recent information sources.

D. Benchmark 3: Existing Metalinguistic Evaluation (Beguš et al. Dataset)

- Utilize the 120-item open-source dataset provided by Beguš et al. (2025), which prompts models to "analyze ambiguous structures, linguistic recursion, syntactic movement, and phonological generalization" (p. 3456).
 - This strictly tests the models' theoretical linguistic capabilities without the time commitment of designing a custom benchmark from scratch.
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IV. Expected Results and Analysis

A. Performance Metrics and Rank Stability

- Compare accuracy percentages across the three benchmarks. Use statistical ranking correlation to determine if the best model on the static MMLU remains the best on LiveBench and the metalinguistic tasks.

B. Analysis of Metalinguistic Capabilities

- Synthesize the results from the metalinguistic dataset. Analyze whether models exhibit iterative reasoning and generalized abstraction when processing complex linguistic formalisms, such as drawing syntactic trees or applying phonological rules (Beguš et al., 2025).

C. Evidence of Data Contamination Impacts

- Analyze the performance delta between static MMLU scores and dynamic LiveBench scores. A significant drop in relative performance on dynamic tasks may serve as empirical evidence of overfitting or test set leakage in static environments (White et al., 2024).

V. Conclusion

A. Summary of Findings

- Provide a synthesis of how dynamic and metalinguistic evaluations expose the true generalization capabilities of LLMs compared to static benchmarks.

B. Implications for Computational Linguistics

- Discuss how the integration of formal linguistic tasks provides a more rigorous metric for understanding models' true capabilities, proving that "metalinguistic ability, one of the few uniquely human properties of language, can emerge in artificial neural models" (Beguš et al., 2025, p. 3464).

C. Future Work

- Recommend the continued development of living benchmarks, such as LiveBench, to keep pace with rapid LLM advancements and avoid the pitfalls of test set contamination (White et al., 2024).
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VI. Bibliography

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